

Abubakar Sadiq Shittu

📍 Knoxville, Tennessee, United States

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🏠 Internship Availability: May 2027 – August 2027

🎓 EDUCATION

University of Tennessee Knoxville, United States.

01/25–05/28 (Anticipated)

Ph.D. Computer Science

GPA: 3.67/4.00

Selected Coursework: Software Engineering, Algorithms, Systems Organization, Data-Driven Security, Digital Archaeology

Advisor: [Dr. Scott Ruoti](#)

Newcastle University, Newcastle upon Tyne, United Kingdom.

09/23–09/24

M.Sc. Advanced Computer Science

GPA: 4.00/4.00

Selected Coursework: Machine Learning, Secure Software Development, Systems Security, Risk & Trust Management

Advisor: [Prof. Thomas Gross](#)

Fountain University Osogbo, Nigeria.

03/18–06/21

B.Sc. Computer Science, First Class Honours

GPA: 4.63/5.00

Advisor: [Dr. Mohammed Lawal](#)

🏢 EXPERIENCE

Graduate Research Assistant

Knoxville, Tennessee, United States

Usable Security and Engineering Research Laboratory

01/25–Present

- Conducted mixed-methods user studies ($n > 20$ per study) on cryptographic key management [1, 2], analyzing qualitative self-reports, log-derived behavioral data, and standardized usability metrics (ASQ, SUS) to identify usability barriers and examine users' experiences with and understanding of the underlying security models, while applying inferential statistics to look for patterns and determine the statistical significance of usability differences across user groups.
- Built and deployed a Docker-based pipeline, orchestrating asynchronous tasks with Celery, collecting user-profile and commit metadata through GitHub's REST and GraphQL APIs, and storing millions of records in PostgreSQL to support subsequent large-scale analysis [3].

Information Technology Helpdesk Intern

Nigeria

Nigerian National Petroleum Corporation

03/19–08/19

- Supported more than 100 users as a primary technical contact, resolving at least five issues daily and maintaining operational continuity.

📖 PUBLICATIONS

- Abubakar Sadiq Shittu, Clay Shubert, John Sadik, and Scott Ruoti. "Setting up TLS authentication was hell": A Usability Study of Client Certificate Authentication. **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26)*, November 15–19, 2026, The Hague, Netherlands. Preprint: [arXiv:2604.14330](#).
- Abubakar Sadiq Shittu, John Sadik, and Scott Ruoti. "A Multi-Month Study of Git Commit Signing." **To appear in** *Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security (CCS '26)*, November 15–19, 2026, The Hague, Netherlands. Preprint: [arXiv:2608.29283](#).

Manuscripts Under Review

- Abubakar Sadiq Shittu, John Sadik, Farzin Gholamrezae, and Scott Ruoti. "Analysis of Commit Signing on GitHub." Manuscript under review, 2026. Preprint: [arXiv:2604.14014](#).

📁 SELECTED CLASS PROJECTS

Machine Learning Pipelines

Newcastle University

- Developed classification, regression, NLP-clustering, and LSTM pipelines across FARS, bacterial-growth, news, and Raspberry Pi sensor datasets using scikit-learn, SMOTE, and Matplotlib.

Advanced RAG Pipeline

University of Tennessee, Knoxville

- Built a RAG question-answering pipeline with LangChain and Gemini by scraping university course schedules with BeautifulSoup, cleaning and storing the data as JSON, chunking documents with RecursiveCharacterTextSplitter, and indexing VertexAIEmbeddings in a vector store for semantic retrieval.

LLM-Based Assistant for Dog Care Guidance

University of Tennessee, Knoxville

- Curated a veterinary-verified dataset of 260 dog breeds and evaluated a TinyLlama-FAISS RAG pipeline against a fine-tuned LLaMA-3.2-1B model on 15 expert-verified queries.

... OTHERS

- Technical Skills:** Python (pandas, NumPy, scikit-learn, Matplotlib, statsmodels), R, C/C++, Docker, SQL, PostgreSQL, REST/GraphQL APIs, Bash, Git, SonarQube, Wireshark, NVivo, MAXQDA, macOS, Windows, Linux/Unix.
- Selected Awards:** **Travel Grant**, *Summer School on Real-World Crypto and Privacy, Croatia* (2024) | **Chevening Scholarship**, *UK Govt. Fully Funded* (~\$70k: tuition, stipend, travel, 2023).
- Professional Service:** Reviewer for *Journal of Cybersecurity* (Q1 SJR, 2025).
- Affiliations & Community:** Graduate Student Member, IEEE Computer Society | [Tennessee Extension Volunteer](#).